

PHARMACOEPIDEMIOLOGY, CHAPTER 26

Comparative Effectiveness Research

A Plain-Language Guide

Why comparing treatments head-to-head in real-world patients is different from testing them in a trial, and how researchers, regulators, and payers around the world try to get it right.

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Based on Setoguchi and Wong, “Comparative Effectiveness Research,” in Pharmacoepidemiology, 6th Edition

What Is Comparative Effectiveness Research?

Comparative effectiveness research (CER) compares the benefits and harms of different ways to prevent, diagnose, treat, or monitor a health condition — testing options directly against each other, in the kinds of patients and settings seen in everyday practice, rather than only against placebo or under ideal trial conditions.

The goal is practical: give patients, clinicians, and the people who pay for care the real-world evidence they need to choose wisely among the options actually available to them.

Efficacy Versus Effectiveness

Efficacy asks whether a treatment works under ideal, tightly controlled conditions — a randomized trial with strict eligibility criteria and close monitoring. Effectiveness asks a harder question: does it still work once real patients take it, alongside other conditions, other medications, missed doses, and everyday life?

The same glucose-lowering drug may achieve excellent control in healthier, younger trial participants — good efficacy. But in older patients with heart failure, mild fluid retention from the drug can trigger nonadherence or early discontinuation, so it may never achieve good real-world glucose control at all. That gap between the two is exactly what CER is built to study.

CER and Pharmacoepidemiology

The two fields overlap heavily: both lean on real-world data and observational methods to study how treatments actually perform outside the trial setting. CER is broader in scope, spanning drugs, medical devices, diagnostic tests, behavioral and social interventions, and healthcare delivery systems, while placing extra emphasis on head-to-head, best-in-class comparisons.

But CER is also narrower in another sense: it concentrates on real-world, later-stage questions, and especially on head-to-head comparisons among practical alternatives, rather than spanning the full bench-to-bedside research pathway.

A Brief History of U.S. CER Policy

1972 to 1989

The Congressional Office of Technology Assessment (1972), then the National Center for Health Care Technology (1978 to 1982), then the founding of AHRQ (1989).

2010

The Affordable Care Act establishes PCORI, funded by a \$4 billion Patient-Centered Outcomes Research Trust Fund.

2009

The American Recovery and Reinvestment Act commits \$1.1 billion to comparative effectiveness research.

By Nov. 2017

PCORI has disbursed more than \$2 billion across roughly 600 research projects.

Many Definitions, One Idea

Major health organizations — the US Congressional Budget Office, the Institute of Medicine, the US Federal Coordinating Council, PCORI, the UK's NIHR, the European Commission, and international HTA networks — have each published their own formal definition of comparative effectiveness research.

Strip away the differences in wording, and they converge on the same idea: comparing two or more ways of managing a condition to find out which works best, for whom, under real conditions of use.

DEFINING CER, 2 OF 5

CER Versus Health Technology Assessment

Health technology assessment (HTA), the term used through much of the rest of the world, is a broader umbrella than CER. It asks the same comparative-effectiveness questions, but it also folds in formal cost-effectiveness analysis — putting a value on the health gained for the money spent.

In practice, many countries route their comparative effectiveness questions through a national HTA process rather than a separately named CER program.

Three Components of CER

The field breaks down into three connected pieces of work, and a full CER program needs all three.

Generation

Carrying out new research that directly compares treatment options.

Synthesis

Pulling together and evaluating the evidence that already exists.

Dissemination

Getting the findings to the patients, clinicians, and policymakers who need them.

DEFINING CER, 4 OF 5

Key Attributes: Real-World and Head-to-Head

Good CER studies real-world patients in real-world settings, not the narrower population enrolled in a typical trial. And it favors head-to-head comparisons between active treatments, rather than comparisons against placebo or no treatment at all.

A landmark 2007 Institute of Medicine report found that less than half of the care delivered in the US is actually supported by solid evidence — partly because so many treatments have only ever been tested against placebo, never against each other.

Stakeholders, Outcomes, and Heterogeneity

Stakeholder Involvement

Patients, clinicians, caregivers, payers, policymakers, and product makers can all help shape the research, though engagement is far more common when setting priorities than during implementation or dissemination.

Effectiveness and Safety

Both benefit and harm are measured, using outcomes that matter to patients rather than lab-value surrogates alone.

Heterogeneity

Good CER looks for whether an effect differs across patient subgroups, and increasingly looks ahead to phenotype and genotype for more individualized prediction.

CER in the United States

In the US, comparative effectiveness research grew out of decades of federal policy experiments, and today runs largely through PCORI, a nonprofit authorized by Congress and funded by a dedicated trust fund rather than ordinary annual appropriations.

PCORI's mandate is explicitly patient-centered: it funds research chosen with patient and stakeholder input, on outcomes that matter to patients, and by law it cannot use cost-per-quality-adjusted-life-year thresholds to judge a treatment's value.

CER in Europe

Across Europe, the broader term health technology assessment, or HTA, does the work that CER does in the US — and goes further, since HTA already builds in formal cost-effectiveness analysis rather than treating cost as a separate question.

The United Kingdom's National Institute for Health and Care Excellence, or NICE, created in 1999, became one of the world's most influential and closely studied HTA models.

NICE's Four Kinds of Guidance

NICE Guidelines

Broad recommendations for clinical practice across a condition or care pathway.

Technology Appraisals

Rulings on whether a specific product, such as the cancer drug ixazomib, should be funded.

Medical Technologies and Diagnostics

Evaluations of devices and tests, such as the Neuropad diagnostic patch, which NICE found was not well supported by evidence.

Interventional Procedures

Guidance on procedures such as intravesical microwave hyperthermia for bladder conditions.

Advisory Versus Regulatory Roles

Countries differ sharply in how much authority their HTA bodies actually hold. France, Germany, and the Netherlands use HTA findings mainly in an advisory role, informing decisions made elsewhere in the system.

The UK, Denmark, and Sweden go further, giving their HTA bodies real regulatory authority: their recommendations directly determine what gets reimbursed. A five-country comparison also found real differences in how independent and how multidisciplinary these committees are.

Hong Kong: A Case Study

Bloomberg ranked Hong Kong's healthcare system the most efficient in the world in 2017 to 2018. Yet Hong Kong runs parallel public and private healthcare sectors that do not share data or treatment evaluations with each other.

Despite its strong outcomes, Hong Kong has not adopted a formal, system-wide comparative effectiveness research process — a reminder that efficiency and formal CER infrastructure do not always travel together.

CER in Africa and Asia

South Africa has published a ten-year white paper laying out a national vision for building health technology assessment capacity from the ground up.

China, India, and South Korea each take notably different approaches to comparative effectiveness work, reflecting real differences in how their healthcare systems are organized and where their institutional priorities lie.

THE SCOPE OF CER, 1 OF 3

What CER Covers

CER's scope is broad by design. It spans drugs, medical devices, diagnostic tests, behavioral and social interventions, and healthcare delivery and policy systems — anywhere a real choice exists between two or more ways of managing a condition.

It also looks at the full picture of that choice: intended benefits, unintended harms, and the economic implications of each option.

THE SCOPE OF CER, 2 OF 3

The Knowledge Gaps CER Can Fill

Randomized trials remain the traditional evidence standard, but they leave real gaps. Many compare a drug only to placebo, not to the other active treatments a clinician is actually choosing between.

Trial populations are often narrower and healthier than typical patients, thanks to strict eligibility criteria, and follow-up is often short, relying on surrogate endpoints rather than outcomes patients actually care about, like hospitalization or quality of life. CER is built to fill exactly these gaps.

The Three Goals of CER

Inform Real-World Decisions

Inform health care decisions about both the intended benefits and the unintended harms of an option, in real-world use.

Put New Options in Perspective

Put new technologies and treatments into proper perspective relative to older, already-established ones.

Identify Who Responds

Identify which patients are more, or less, likely to respond to a particular intervention.

Together, these goals point at the same outcome: care that is higher quality and better value, chosen with real evidence.

Systematic Reviews and Meta-Analyses

Two tools anchor evidence synthesis in CER: a systematic review comprehensively and transparently searches out all the relevant existing studies, while a meta-analysis statistically pools their results into a single combined estimate.

Both are guided by established reporting standards, such as those from the Cochrane Collaboration and the PRISMA guidelines, to keep the process transparent and reproducible.

When Meta-Analyses Mislead

Meta-analyses are more subjective than they look. Every step — which studies to include, how to define the outcome, which statistical model to use — is an investigator judgment call, and Simmonds and colleagues found that expert reviewers analyzing the very same data can reach meaningfully different conclusions.

Publication bias compounds the problem, since studies with positive findings are more likely to be published at all. And pooling from individual patient-level data, when it is available, gives a more reliable answer than pooling from published summary statistics alone.

EVIDENCE SYNTHESIS, 3 OF 3

The Case for Observational Meta-Analyses

Observational studies are central to CER: they capture routine, real-world practice, and can study larger populations for longer periods than most trials manage. Most CER studies, historically, have used observational designs.

That realism comes with a tradeoff, and it shows up sharply in meta-analysis: whether observational studies even belong in a pooled analysis is genuinely debated, since their designs and populations vary more than trials do. Man and colleagues pooled observational data and found methylphenidate use was associated with fewer injuries; Kirtane and colleagues showed a case where analyzing studies separately was the more informative choice.

More precision does not rescue biased studies. And a systematic review is a living evidence summary, not a permanent answer — it has to be updated as new evidence emerges to stay useful.

Confounding by Indication

Doctors do not prescribe at random. The very reason a patient is offered — or denied — a treatment is often tied to their underlying risk, and that risk gets tangled up with any apparent effect of the treatment itself.

This bias tends to be worse when studying a drug's intended effect than its unintended, side effects — because the intended effect is precisely what shaped the prescribing decision in the first place.

CONFOUNDING BY INDICATION, 2 OF 4

When Indication Hides in Plain Sight

Sometimes confounding by indication is obvious: patients given a drug for severe disease are sicker to begin with. Other times it is much subtler.

Physicians choosing between an ACE-inhibitor and a thiazide diuretic for hypertension may be responding to small differences in kidney function or diabetes status that independently affect outcomes. Patients selected for methotrexate plus a TNF-inhibitor in rheumatoid arthritis often have more severe or treatment-resistant disease than those kept on methotrexate alone.

Methods That Help, and Their Limits

Statistical adjustment can help: propensity-score methods, disease-risk scores, and doubly-robust methods that combine two approaches for extra protection. Design-based approaches offer another route: restricting the study population, comparing two active treatments instead of a treatment to non-use, or using self-controlled designs that compare each patient only to themselves.

Most of these methods can only adjust for confounders that were actually measured and recorded. Instrumental-variable methods are a partial exception: under strong assumptions, they can help address confounding that was never measured at all — but finding a genuinely valid instrument is exceptionally difficult.

The Carotid Stenting Example

The chapter's authors compared carotid stenting to carotid endarterectomy — two ways of treating narrowed neck arteries — using insurance claims data.

Claims data alone was not enough to fully account for confounding by indication. Drawing a confident conclusion needed richer clinical detail than administrative claims typically capture.

HEALTHY USER BIAS, 1 OF 3

The Healthy User and Candidate Effect

People who take preventive medications as recommended — statins, hormone replacement therapy, antihypertensives — also tend to eat better, exercise more, and use more preventive care overall.

A related pattern, healthy-candidate bias, comes from the clinician's side: doctors tend to prescribe preventive drugs to patients who already look robust enough to benefit from and tolerate them. Either way, some of the apparent drug benefit may really belong to these other, unmeasured behaviors.

HEALTHY USER BIAS, 2 OF 3

A Falsification Test: ICDs in Heart Failure

The chapter's authors tested for hidden bias using a falsification outcome: an outcome that the treatment being studied has no plausible way to affect.

Looking at implantable cardioverter-defibrillators in heart failure patients, they found improvement even on outcomes the device could not plausibly influence — a red flag that unmeasured healthy-user or healthy-candidate bias, not a true causal effect, was doing some of the work.

Why This Bias Is So Hard to Fix

The factors that separate adherent, health-conscious patients from everyone else are numerous, and most of them never make it into a research database: general health-seeking behavior, diet, exercise, functional status, and use of other preventive care.

The sobering finding: healthy-user and healthy-candidate bias can be as large as, or even larger than, the true treatment effect being studied — and unlike confounding by indication, it may resist even the best statistical fixes, because so little of it is ever actually measured.

Data Sources for CER

CER draws on two broad kinds of data, grouped in a way that may surprise you: medical and pharmacy records, electronic health records, and newly collected study data all count as primary sources.

Administrative claims and clinical registries, by contrast, count as secondary sources, since they were assembled for billing or monitoring rather than for the study at hand. The best source depends on the question, and on how much it needs both validity and generalizability.

DATA AND LINKAGE, 2 OF 4

Record Linkage Versus Multidatabase Studies

Two different strategies exist for combining data sources. Record linkage connects an individual person's records across different databases, such as tying pharmacy records to hospital records for the same patient.

Multidatabase studies instead run the same analysis separately across several databases and then compare or combine the results, without ever linking individual records. Lichtenstein and colleagues used record linkage to connect ADHD medication records to criminal-justice records, studying whether medication use is associated with a person's risk of criminal behavior.

Linkage does quieter work too: connecting claims to clinical registries can validate claims-based outcome measures, and connecting trial participants to claims can extend long-term follow-up well beyond a trial's original observation window, at low added cost.

The Two Challenges of Linkage

Linking records across systems raises two distinct kinds of challenges. Methodologic challenges are about accuracy: making sure the linkage is correct and that variables mean the same thing across different sources.

Ethical challenges are about privacy: how comfortable patients actually are with their data being connected across systems. The chapter's authors ran their own patient survey on this question, echoing a similar survey by the Wellcome Trust in the UK.

Multidatabase Networks and Variability

Distributed-network approaches let researchers study multiple databases together without ever centralizing the raw data. One approach standardizes every site's data into a shared common data model.

Another, used by the Canadian Network for Observational Drug Effect Studies, instead has every site run an identical shared protocol on its own local data. Either way, a persistent challenge remains: telling true variation in findings across sites apart from variation that is really just a technical or database-specific artifact.

Pragmatic Trials

A pragmatic trial is a randomized trial deliberately designed to look and feel like real-world practice: broad, inclusive eligibility criteria, and active treatments compared against other active treatments rather than against placebo.

The aim is to keep randomization's power to rule out confounding, while still producing results that generalize to the everyday patients and everyday decisions clinicians actually face.

EXPERIMENTAL CER, 2 OF 4

ALLHAT and ZODIAC

Pragmatic trials and cluster-randomized trials are both examples of “large, simple” CER designs built to test real treatment choices at enormous scale.

ALLHAT

About \$120 million, roughly 42,000 patients, and 600 sites, comparing blood-pressure-lowering drugs head-to-head.

ZODIAC

About \$85 million, roughly 18,000 patients, across 18 countries.

CER often searches for smaller differences than older studies did, because it compares effective treatments head-to-head in broad, heterogeneous populations — dramatic, transformational effects have grown rarer, and effect sizes have shrunk even in placebo-controlled trials.

The Cost Problem, and Point-of-Care Trials

Trials at ALLHAT's and ZODIAC's scale are extremely expensive, and even well-run ones face loss to follow-up and, often, a lack of blinding.

A lower-cost alternative is the point-of-care, or electronic, trial, embedded directly into routine care through existing electronic health record systems — piloted using the UK's General Practice Research Database and the US Veterans Affairs EHR.

Success takes three things: a large enough patient population reachable through the record system, workflows that support randomization without disrupting routine care, and clinicians and patients willing to accept randomization in routine care when the treatment options are genuinely in equipoise. UK experience showed substantial practice and clinician attrition during the trial's setup process.

Choosing Experimental or Observational

Four considerations guide the choice: is it ethical to randomize patients to one option or the other; what would each approach cost; could an observational design answer the question efficiently enough; and would the bias in an observational study be so severe and intractable that only a trial can answer it credibly.

Bias is a matter of degree, not a simple yes-or-no switch. Even an observational study with real limitations still adds value when it is reported fairly and honestly about what those limitations are.

Why Evidence Alone Isn't Enough

A systematic review of published CER guides found more than 300 recommendations across just nine documents — a sign of how much thought has gone into doing comparative effectiveness research well.

But good evidence sitting in a journal article does not automatically change what happens in the exam room. Getting findings into practice is a distinct challenge, with its own science.

Four Goals of Dissemination

Synthesis

Pulling together what is already known into a clear, usable summary.

Guideline Translation

Converting research findings into clinical practice guidelines, then updating those guidelines as evidence changes.

Knowledge Exchange

Putting guidelines to work through order-entry prompts, reminders, clinician education, and decision-support tools.

Monitoring

Tracking whether dissemination efforts are actually changing what happens in practice.

DISSEMINATION, 3 OF 4

When Good Evidence Doesn't Change Practice

Prostate-specific antigen, or PSA, screening is a telling case: evidence of limited benefit and real harms took years to meaningfully shift widespread screening habits.

ALLHAT itself tells a similar story. Despite strong trial evidence favoring a specific, inexpensive class of blood-pressure drugs, real-world prescribing patterns were slow to change even after the results were published.

DISSEMINATION, 4 OF 4

Five Reasons Practice Doesn't Change

Timbie and colleagues traced the evidence-to-practice gap to five root causes.

Financial Incentives

Payment structures that don't reward the evidence-based choice.

Ambiguity or Doubt

Lingering questions about whether the evidence really applies, or is valid.

Cognitive Biases

Confirmation bias, and a pull toward newer interventions and technology over older, proven ones.

Unmet End-User Needs

Dissemination efforts that don't address what clinicians and patients actually need from the evidence.

Underused Decision Support

Clinical decision-support tools exist but are too rarely built into everyday workflows.

A related finding: pay-for-performance programs tend to improve documented process measures more than they improve actual patient outcomes.

CER Is Not About Cutting Costs

A deliberate design choice runs through the field: comparative effectiveness research exists to produce better-informed care, not to shrink health care spending.

PCORI, for instance, is barred by law from using cost-per-quality-adjusted-life-year thresholds to judge a treatment's value. CER is not meant to be used as a cost-containment tool.

CER can just as easily support a more expensive option, if the evidence shows it is genuinely better. The goal is identifying better care, not automatically cheaper care.

The Funding Landscape

In the US, three main sources fund comparative effectiveness work: PCORI, supported by its dedicated trust fund; the Agency for Healthcare Research and Quality; and other programs across the Department of Health and Human Services.

Funding is not spread evenly. It varies by disease area and by population, with certain conditions and patient groups receiving considerably more research attention and investment than others.

How Much CER Is Actually Happening?

Despite the funding and policy attention, true CER remains a minority share of the research being done.

Hochman and McCormick estimated that only about one-third of medication studies actually qualify as comparative effectiveness research. Separately, only about 22% of studies registered on ClinicalTrials.gov were classified as comparative-effectiveness studies.

The Cost of U.S. Health Care

The stakes behind all of this are enormous. US healthcare spending totaled about \$3.3 trillion in 2016, and is projected to reach roughly \$5.7 trillion by 2026.

CER will not shrink those numbers directly — that was never its job. But getting more of that spending to go toward care that is genuinely proven to work is exactly the problem it is built to help solve.

THE FUTURE OF CER, 1 OF 3

The Future of CER

Three threads run through where the field is headed next: real-world evidence is becoming a formally defined and widely used concept in its own right, industry interest in it is rising, and the field still needs to build up the specialized workforce capable of doing this work well.

None of these are finished projects — they are the next chapter of the same effort this chapter describes: comparing real options, honestly, for real patients.

Real-World Evidence and Industry

Real-world evidence, or RWE, has become its own defined term, with formal definitions now published by the FDA and by the European Union's GetReal Glossary.

Pharmaceutical industry interest in RWE and CER is rising too, illustrated by industry-funded analyses comparing different direct oral anticoagulants head-to-head in real-world use.

Building the Workforce

Growing this field takes trained people, not just funding for studies.

PCORI has invested about \$30 million, and AHRQ about \$94 million, specifically in training the next generation of comparative effectiveness researchers.

That workforce must span trial design, observational methods, evidence synthesis, data linkage, and implementation — and work effectively across clinical and methodological disciplines.

Medicine Is Still an Art

Sir William Osler described medicine as “an art, based on science.” That description frames how this chapter closes: comparative effectiveness research strengthens the science, but it does not replace clinical judgment.

Better evidence is not the same thing as a decision already made. Someone still has to apply it to the patient actually sitting in front of them.

The Limits of Evidence

Kerridge and colleagues make the point that medical decision-making draws on a broad spectrum of knowledge, not evidence alone — clinical experience, the individual patient's circumstances, and their own values and preferences all matter too.

Two opposite risks sit on either side of this: therapeutic nihilism, distrusting all treatment because no evidence is ever perfect, and overly rigid, algorithmic care that mechanically applies population-level findings to every individual patient. Subgroup analysis helps, but it has real limits in how far it can go toward truly individualized answers.

Tolerating Uncertainty

Practice variation — different clinicians managing similar patients differently — is often treated purely as a problem to be stamped out.

The chapter closes by suggesting it can also be tolerated, even encouraged, as a source of real-world natural experiments — exactly the kind of comparative data that will fuel the next generation of comparative effectiveness research.

WRAP-UP, 1 OF 2

The Throughline

Comparative effectiveness research exists because knowing a treatment works in a trial is not the same as knowing it works for the real patients, and the real choices, clinicians actually face. Getting there means confronting hard biases — confounding by indication, healthy-user effects — head-on, not wishing them away.

Good evidence, honestly generated, is still only half the job. Getting it into practice, funding it fairly, and remembering that medicine stays part art even as the science improves, is the other half.

Thank You

Comparative effectiveness research sits at a genuinely hard crossroads: it must be rigorous enough to correct for the biases that plague real-world data, humble enough to admit what it still cannot answer, and practical enough that its findings actually reach the people making decisions.

No single study, database, or country's policy model has this fully solved. That is precisely why the field keeps building better methods, better data infrastructure, and better ways of closing the gap between evidence and everyday care.